Why Copy-Paste Augmentation Hurt Detector Performance

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Abstract

This report examines why copy-paste augmentation reduced detector performance on MinneApple and Global Wheat Head. We compare random and score-guided copy-paste, validate the scoring signal against miss and false-positive behavior, and inspect the resulting feature space with frozen YOLOv8s embeddings. Across both datasets, the augmentation policies move the training distribution but do not move it closer to the test set in a way that improves generalization. The evidence supports a simple conclusion: harder-looking samples are not automatically better training samples, and score guidance alone is not enough to make copy-paste safe.

1 Introduction

1.1 Problem

Copy-paste augmentation was introduced to increase object diversity, but in this project it consistently lowered performance on the original test sets. The failure is important because the augmentation is not random noise; it is a structured intervention that should, in principle, improve coverage of hard examples.

1.2 Hypothesis

The working hypothesis is that the selected pasted objects are difficult under the current model, but not necessarily aligned with the target test distribution. If that is true, then the augmentation can raise sample difficulty while still hurting detector calibration and localization quality.

1.3 What Is Being Tested

I compare baseline training against two copy-paste policies: random selection and score-guided selection through MinImageScorer. The goal is not just to see whether performance changes, but to check whether score guidance changes the type of samples the model sees in a useful way.

1.4 Scope

The report focuses on two datasets with different failure modes. MinneApple uses same-image copy-paste, while Global Wheat Head uses cross-image copy-paste across a more heterogeneous distribution. That contrast makes it possible to separate augmentation design issues from dataset-specific effects.

2 Related Work

2.1 Copy-Paste Augmentation

Copy-paste augmentation is usually used to increase instance count and improve exposure to rare objects. Prior work shows that it can help when pasted objects remain plausible in context, but it becomes unreliable when object placement or scene composition breaks the underlying data distribution.

2.2 Hard-Sample Mining

Score-based sampling and hard-example mining are attractive because they target the examples the model currently struggles with. The risk is that difficulty is a training signal, not a guarantee of usefulness, so the selected samples may over-emphasize atypical or noisy cases.

2.3 Domain Shift in Detection

Object detectors are sensitive to background, scale, overlap, and scene statistics. If augmentation changes those factors too aggressively, the model can fit the training mixture while moving away from the evaluation domain.

2.4 Feature-Space Diagnosis

Feature-space analysis is a useful sanity check when metrics alone are ambiguous. Image embeddings from a frozen backbone can show whether augmented samples stay near the source manifold, collapse into a narrow region, or move farther from test than the original data.

3 Methodology

3.1 MinImageScorer

The selection policy is driven by MinImageScorer, which ranks images and objects by difficulty rather than by frequency. The intent is to bias paste selection toward samples the current detector finds weak, including low-confidence detections and missed objects.

3.2 Random and Score-Guided Selection

Two augmentation policies are compared throughout the report. Random copy-paste samples candidate objects without any score preference, while score-guided copy-paste prefers high-scoring objects according to the scoring model. This isolates the effect of the selection rule itself.

3.3 Boundary Conditions

The final comparison uses the following constraints:

Setting	Value	Why it matters
Dataset ratio	0.3	Keeps the augmented set large enough to matter without overwhelming the original distribution.
Minimum objects/image	2	Avoids trivial pasted images with too little content.
Maximum objects/image	5	Limits clutter and over-composition.
Scale range	0.8–1.2	Keeps pasted objects close to their natural size.
Rotation range	$\pm 15°$	Adds variation without making the object implausible.
Overlap handling	avoid overlap = true	Prevents pasted objects from colliding with existing instances.
Max object reuse	4	Reduces repeated copying of the same instance.
Score weighting	linear	Keeps the selection rule interpretable.
Alpha	1.0	Uses a conservative score bias rather than an aggressive one.

3.4 Analysis Pipeline

The report uses two analysis layers. First, it measures detector performance on the original test set using COCO-style metrics. Second, it extracts frozen YOLOv8s image embeddings from the P5 feature map, applies global average pooling, and compares the centroid distance between training, augmented, and test samples.

3.5 Why This Design

This setup is deliberately simple. It keeps the control logic local to the selection policy, so when performance drops it is easier to tell whether the failure comes from the augmentation rule, the pasted content, or a mismatch between the training and test distributions.

4 Experimental Setup

4.1 Datasets

The experiments use MinneApple and Global Wheat Head in YOLO format. The train/val/test splits are fixed, and all reported test metrics are computed on the original test split rather than on augmented data.

4.2 Split Summary

Dataset	Split	Images	Objects	Notes
MinneApple	Train	536	22,815	73.61% small, 26.39% medium
MinneApple	Val	134	5,367	Same annotation format as train
MinneApple	Test	331	12,285	Held out for final evaluation
Global Wheat Head	Train	2,700	115,777	Dense, overlapping heads
Global Wheat Head	Val	337	14,622	Different image domains
Global Wheat Head	Test	339	15,012	Held out for final evaluation

4.3 Model and Training

All detector runs use YOLOv8s with COCO-pretrained weights. Augmentation is disabled during detector training so that the effect of the copy-paste pipeline remains visible, and the model is trained for 100 epochs with early stopping where applicable.

4.4 Evaluation Protocol

Every comparison is evaluated on the original test set. The report tracks COCO AP, AP50, AP75, and size-specific AP where available. For the feature-space analysis, embeddings are computed from a frozen backbone so that the geometry reflects sample structure rather than detector adaptation.

5 Results and Analysis

5.1 Main Performance Table

MinneApple stays below baseline in both augmentation modes, and Global Wheat Head shows the same pattern. The exact values are below.

MinneApple	**COCO AP**	**AP50**	**AP75**	**AP**$_{small}$	**AP**$_{medium}$	**AP**$_{large}$
Baseline	0.4387	0.7708	0.4519	0.2996	0.5922	0.6449
Random copy-paste	0.4152	0.7409	0.4273	0.2792	0.5633	0.6689
Score-guided copy-paste	0.4277	0.7592	0.4405	0.2856	0.5809	0.6374

Global Wheat Head	**COCO AP**	**AP50**	**AP75**	**AP**$_{small}$	**AP**$_{medium}$	**AP**$_{large}$
Baseline	0.5070	0.9223	0.4907	0.1365	0.5013	0.5465
With-score copy-paste	0.4837	0.9045	0.4549	0.1090	0.4772	0.5370
Without-score copy-paste	0.4857	0.9048	0.4556	0.1182	0.4796	0.5288

5.2 Score Validation

The score is meaningful, but it is not sufficient. In MinneApple, the score correlates strongly with miss and false-positive behavior, with Pearson/Spearman values around 0.6978/0.6883 and 0.6788/0.6983. In Global Wheat Head, the same trend is stronger for some cases, with values around 0.8158/0.8180 and 0.7180/0.7157. That means the score is a valid hardness signal, but not a guarantee that the selected samples will improve test alignment.

5.3 Visual Comparison

The augmented-vs-base comparison below is the same style used in the notebook viewer. It is useful as a fast qualitative check because it exposes obvious placement, clutter, and context problems.

5.4 Dataset Diagnostics

MinneApple is dominated by small apples: 16,793 small objects and 6,022 medium objects in the training split. That bias helps explain why copy-paste can increase clutter without producing a more useful target distribution. Global Wheat Head is even more sensitive because it contains 115,777 training instances spread across more heterogeneous scenes and stronger domain variation.

5.5 Feature-Space Analysis

The feature-space notebook confirms the same story. On MinneApple, the augmented samples stay close to the original train manifold while the test set remains separated. On Global Wheat

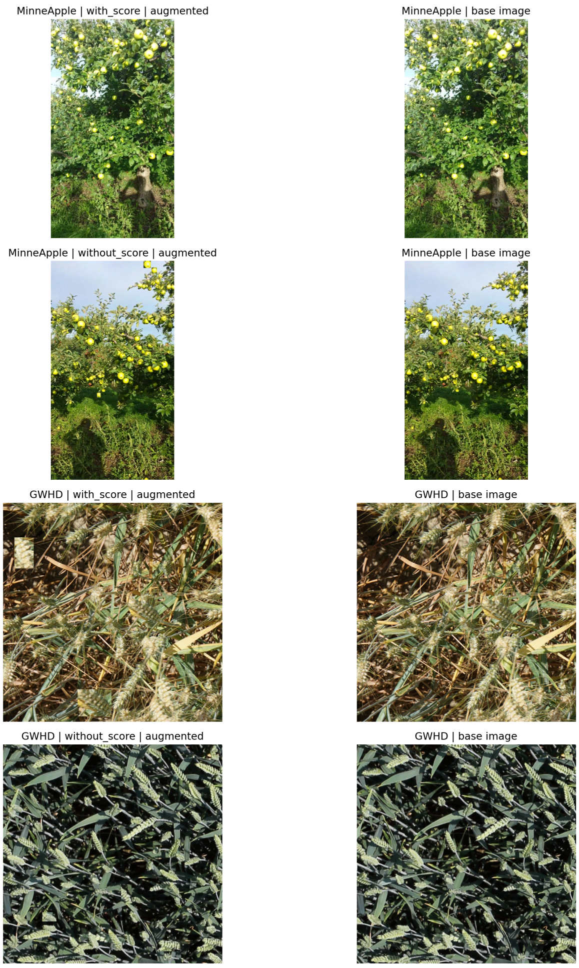

Figure 1: Representative augmented-vs-base comparison for MinneApple and Global Wheat Head.

Head, the centroid distances are 16.8058 for original train to test, 16.2602 for random augmentation to test, and 17.3665 for score-guided augmentation to test. The score-guided case is the farthest from test, which matches the performance drop.

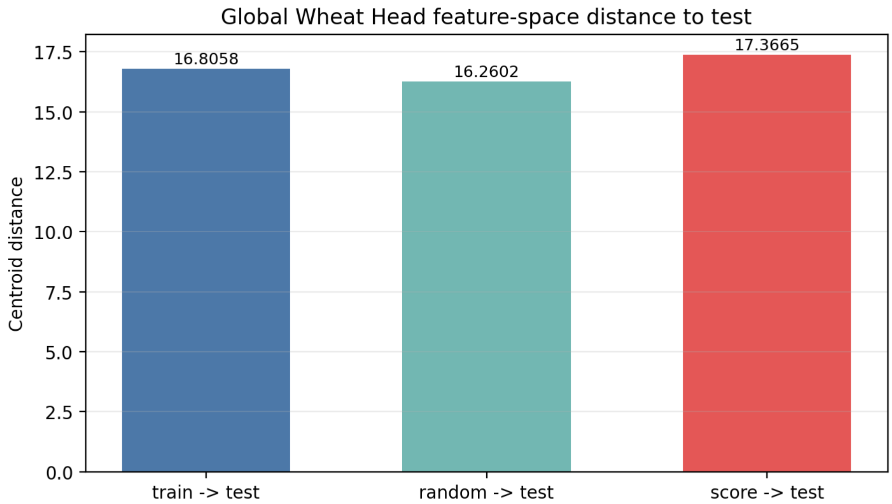

Figure 2: Global Wheat Head feature-space distances from the frozen YOLOv8s embedding.

5.6 Alpha Sweep

The alpha sweep shows that stronger score weighting does not fix the problem. It pushes the augmented distribution harder toward difficult samples, but the test performance still degrades as alpha increases.

Alpha	**Baseline AP**	**Augmented AP**	**Δ**
1.0	0.3844	0.3720	-0.0124
3.0	0.3844	0.3652	-0.0192
5.0	0.3844	0.3619	-0.0225

6 Discussion

6.1 Why the Score Helps and Still Fails

The score does capture a real notion of difficulty, but difficulty alone is too blunt. A hard object can still be a bad training target if it is poorly placed, highly occluded, duplicated too often, or drawn from a scene that does not resemble the test distribution.

6.2 MinneApple vs Global Wheat Head

MinneApple is less pathological because same-image copy-paste stays within the same scene family. Even there, however, the augmentation does not improve the original test result. Global Wheat Head is harder because cross-image pasting mixes domains and makes the distribution shift more visible.

6.3 Boundary Conditions

The current pipeline works only inside a narrow band of settings. If the augmentation ratio is too high, if alpha is too aggressive, or if object reuse is too frequent, the pasted samples start to dominate the training mixture and the model learns a distribution that is too far from the test set.

6.4 Practical Recommendation

The safest conclusion is not to reject copy-paste entirely, but to treat it as a constrained intervention. The policy should be kept conservative, validated with the original test split, and checked with feature-space diagnostics before it is trusted as a training improvement.

6.5 Limitation

The current analysis is image-centric and backbone-centric. It is good enough to explain the broad failure mode, but it does not fully isolate whether the dominant problem is object placement, context mismatch, or scoring bias.

7 Conclusion

The first conclusion is direct: copy-paste augmentation did not improve the detector on either dataset. In both MinneApple and Global Wheat Head, the augmented models stayed below the baseline on the original test set.

The second conclusion is that score guidance is not a cure. It produces a meaningful hardness ranking, but hardness is not the same as usefulness. The score can select difficult objects while still pulling the model away from the test distribution.

The third conclusion is that the augmentation must be constrained more aggressively if it is going to help. The current settings are conservative enough to remain plausible, but not conservative enough to guarantee better generalization.

The final conclusion is practical: the report gives a reproducible explanation for the failure, and the next iteration should optimize for distribution alignment first, then difficulty second.